

Critical Analysis: Can LLMs Express Their Uncertainty? An Empirical Evaluation of Confidence Elicitation in LLMs

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1 Methodological Strengths

1.1 Clear problem decomposition and a compositional experimental framework

The study targets *black-box* confidence elicitation for Large Language Models (LLMs), where only text inputs and outputs are available. A key methodological strength is the decomposition of the design space into three orthogonal components, namely prompting, sampling, and aggregation, followed by systematic evaluation of their combinations. This structure enables controlled comparisons that isolate which component drives improvements in calibration versus failure prediction, rather than attributing performance changes to an opaque end-to-end heuristic.

1.2 Task formulation aligned with deployment requirements

The evaluation is grounded in two operational objectives for confidence scores:

- **Calibration:** confidence values should match empirical correctness frequency, operationalized with Expected Calibration Error (ECE).
- **Failure prediction:** confidence should separate correct from incorrect outputs, operationalized with Area Under the Receiver Operating Characteristic curve (AUROC), and complemented with AUPRC-Positive and AUPRC-Negative to account for class imbalance effects.

This separation is methodologically effective because calibration and discrimination can diverge in practice, and the study explicitly demonstrates failure cases where calibration improves without improved failure prediction.

1.3 Diverse benchmarks and model coverage to probe generality

The study benchmarks across five task types, spanning commonsense reasoning, arithmetic reasoning, symbolic reasoning, professional knowledge, and ethical knowledge, using eight datasets. This design reduces the risk that conclusions are an artifact of a single benchmark family. Model coverage includes multiple widely used LLMs (both proprietary and open-source), enabling analysis of how confidence behavior changes with capability scaling. The inclusion of professional knowledge tasks is particularly valuable because these tasks reveal failure prediction weaknesses that are less visible on arithmetic or simpler reasoning datasets.

1.4 Method space includes human-inspired elicitation protocols beyond a single prompt

Beyond a vanilla prompt that requests an answer and a confidence score, the paper evaluates multiple human-inspired prompting strategies:

- **Chain-of-Thought (CoT):** step-by-step reasoning prior to confidence reporting.
- **Self-Probing:** confidence elicitation in an independent session given a candidate answer.
- **Multi-Step:** per-step confidence elicitation with an explicit aggregation rule.
- **Top- K :** multiple candidate answers with associated probabilities.

This breadth strengthens internal validity by showing that confidence elicitation is not reducible to a single prompt template, and it enables analysis of which elicitation forms reduce overconfidence versus merely increase accuracy.

1.5 Aggregation strategies include formal objectives and interpretable mechanisms

Aggregation is not treated as an ad hoc heuristic. Two key contributions are the explicit definitions of consistency-based confidence and confidence-weighted aggregation, together with a ranking-based aggregator (Pair-Rank) motivated by the empirical claim that relative rankings can be more reliable than absolute verbalized probabilities. The Pair-Rank method is specified via a conditional maximum-likelihood objective under a categorical sampling-without-replacement assumption, and the paper provides a proof sketch for its core probability form. This explicitness supports reproducibility and makes assumptions inspectable.

1.6 Systematic ablations and controlled comparisons

The experimental section is organized to examine components of the framework in isolation:

- Baseline characterization of vanilla verbalized confidence, including distributional diagnostics (confidence mass in 80–100, multiples of 5) and reliability diagrams.
- Prompting-only comparisons on fixed datasets and models to assess elicitation effects.
- Sampling studies that vary the number of responses M and show saturation behavior, paired with explicit cost trade-offs (linear query growth).
- Aggregation studies comparing Consistency, Avg-Conf, and Pair-Rank under a fixed Top- K and sampling configuration.

This sequencing constitutes an ablation-style methodology that clarifies causal contribution paths.

1.7 Transparency and reproducibility measures

Several reporting practices support reproducibility:

- Public release of code.
- Explicit prompt templates summarized in the main text, with complete prompts referenced in an appendix.
- Clear specification of evaluated models, datasets, and metrics, with implementation details deferred to an appendix.

While not exhaustive, these measures reduce ambiguity for replication.

2 Key Limitations

2.1 Evaluation scope restricted to single-ground-truth QA settings

The evaluation focuses on fixed-form and free-form question answering tasks where the ground-truth answer is unique. This restriction limits conclusions about confidence elicitation for open-ended generation, such as summarization, dialog, or creative generation, where correctness is multi-dimensional and confidence should incorporate criteria beyond exact matching. The paper explicitly notes this as a limitation and defers such tasks to future work.

2.2 Dataset representativeness and domain shift are not directly tested

The benchmark suite spans several task families, but it does not explicitly test robustness under distribution shift, adversarial query variations beyond the misleading prompt, or real-world deployment distributions where question phrasing, context length, and ambiguity differ from curated benchmarks. Consequently, the reported calibration and discrimination results cannot be assumed to transfer to production settings without additional stress tests.

2.3 Limited reporting of experimental settings affecting replicability and interpretation

Several implementation details are not specified in the main narrative:

- Sampling hyperparameters such as temperature values and decoding settings across experiments.
- Exact evaluation rules for mapping free-form answers to correctness across datasets, especially for numeric or formatted outputs.
- Query budgets and rate-limit considerations for proprietary APIs that influence feasible M .

The absence of these details in the core exposition complicates interpretation of performance differences that can depend strongly on decoding configuration, especially for multi-sample elicitation.

2.4 Confidence reporting format may induce discrete artifacts

Verbalized confidences concentrate in the 80–100 range and frequently appear as multiples of 5, suggesting discretization inherited from training corpora or conversational conventions. This introduces quantization that limits granularity, which can harm both calibration (by forcing coarse bins) and failure prediction (by compressing scores). The paper recognizes the phenomenon but does not introduce a mechanism to correct discretization beyond aggregation.

2.5 Calibration improvements can be conflated with accuracy improvements

A key limitation highlighted empirically is that reduced ECE does not necessarily reflect improved introspection. For instance, on GSM8K under CoT prompting, a highly accurate model can achieve low ECE by assigning uniformly high confidence to all samples, which yields limited or no ranking signal for failure prediction. The study discusses this failure mode, but the evaluation does not separate calibration due to improved confidence estimation from calibration due to improved base accuracy, beyond qualitative interpretation.

2.6 Aggregation evaluation includes a configuration with anomalous discrimination behavior

In the aggregation comparison under Top- K and self-random sampling on GPT-4, Avg-Conf yields high AUROC on several datasets but exhibits notably low AUROC on GSM8K in the reported table. This indicates that aggregation strategies can interact with dataset properties and elicitation formats in non-intuitive ways. The paper motivates Avg-Conf by consistency-driven variance, but the GSM8K result suggests instability that is not fully explained.

2.7 White-box comparison is informative but limited in breadth

White-box baselines are implemented via token probability aggregations (sequence probability, length-normalized sequence probability, and key token probability). The comparison is useful for establishing an upper bound under richer access, but it remains limited to token-probability formulations and does not include stronger semantic uncertainty methods beyond noting computational cost. As a result, the gap between black-box and state-of-the-art white-box uncertainty estimation is not fully characterized.

2.8 Computational cost analysis remains high-level

The study notes that increasing M linearly increases query cost and that performance saturates as M grows, but it does not provide detailed cost accounting (wall-clock time, API cost, or energy). Such reporting is important because the strongest failure prediction improvements rely on multiple samples per query, which may be prohibitive in practice.

3 Technical Bottlenecks

3.1 Semantic uncertainty remains difficult under black-box constraints

Token-likelihood is argued to capture next-token uncertainty rather than semantic truth, motivating verbalized confidence. However, verbalized confidence itself is a generated text field, which inherits the same modeling biases and does not provide direct access to internal epistemic signals. The core technical bottleneck is that black-box outputs compress model uncertainty into a low-bandwidth channel that is entangled with stylistic norms, leading to persistent overconfidence.

3.2 Calibration and discrimination objectives are structurally misaligned

The study demonstrates a fundamental trade-off: a method can reduce ECE while providing weak failure prediction. This is not merely an implementation detail; it arises because calibration metrics depend on average alignment between confidence and accuracy, while discrimination requires per-sample score variability correlated with correctness. Prompting strategies that boost accuracy can reduce ECE even if confidence remains saturated, while variance-based sampling can improve discrimination even if verbalized confidences remain poorly calibrated.

3.3 Variance signals require expensive sampling and remain imperfect

Sampling-based variance is a strong failure prediction signal, but it requires multiple queries and depends on decoding randomness. The technical barrier is twofold:

- **Efficiency:** multi-sample inference incurs linear cost in M .

- **Faithfulness:** sampled diversity depends on decoding hyperparameters, and the sampled set may not faithfully represent the model’s implicit answer distribution under constrained budgets.

The reported saturation with larger M suggests diminishing returns, constraining attainable performance.

3.4 Aggregation is constrained by the reliability of elicited primitives

Consistency-based aggregation is coarse-grained for small M , producing a limited set of possible confidence values. Confidence-weighted aggregation depends on verbalized confidences that are demonstrably overconfident and discretized. Pair-Rank depends on a probabilistic assumption (categorical draws without replacement) that may not match actual decoding dynamics of LLMs, and it introduces an optimization layer whose stability depends on the diversity and coverage of observed rankings. These constraints limit how much aggregation alone can recover reliable uncertainty from noisy elicited signals.

3.5 Professional knowledge tasks reveal a persistent error detection gap

Even when calibration improves to low ECE values, failure prediction remains weak on tasks requiring specialized knowledge. This indicates a bottleneck where models produce fluent justifications and high confidence irrespective of correctness, and where response variance is not sufficiently diagnostic. This challenge is aligned with broader observations about hallucination and self-consistency failures in knowledge-intensive settings.

4 Research Implications

4.1 Current confidence elicitation is insufficient for high-stakes deployment

The empirical overconfidence patterns and AUROC values near the random-guess regime for several settings imply that naive confidence elicitation cannot support reliable risk-sensitive decision making. Confidence outputs that cluster at high values can actively harm downstream systems by suppressing fallback mechanisms or human review.

4.2 Progress in calibration does not imply progress in introspective reliability

The distinction between calibration and failure prediction implies that evaluation of uncertainty methods for LLMs must include both objectives. Relying solely on ECE can misrepresent progress, especially when a method increases base task accuracy and reduces the average confidence-accuracy gap without improving per-instance error detection. This has implications for benchmark design: uncertainty benchmarks should penalize confidence saturation and reward meaningful ranking of failures.

4.3 Black-box uncertainty estimation remains a core open problem

The modest gap between black-box and token-probability white-box baselines suggests that richer access alone does not resolve the uncertainty estimation problem, but the best results remain unsatisfactory. This indicates that the field lacks robust signals that correlate with semantic correctness, both in black-box and in common white-box token-probability approaches.

4.4 Evaluation protocols should reflect cost constraints and operational trade-offs

Sampling and aggregation strategies exhibit measurable improvements but impose higher query cost. Deployment-relevant evaluation should report performance as a function of query budget and explicitly characterize the Pareto frontier between reliability and cost. The paper takes steps in this direction by varying M and recommending a stable configuration, but stronger protocol standardization is needed across the field.

4.5 Connections to broader reliability challenges in generative modeling

The observed overconfidence and weak failure prediction connect to hallucination, self-verification, and calibration challenges in generative models. The results imply that confidence elicitation is not merely a calibration layer; it is entangled with reasoning quality, knowledge grounding, and the model’s ability to represent ignorance. This bridges uncertainty estimation with research on verifiers, retrieval grounding, and self-critique.

5 Potential Research Directions

5.1 Objective-aligned evaluation protocols for uncertainty in LLM outputs

- **Dual-objective benchmarks:** require simultaneous reporting of calibration and discrimination, and introduce metrics that penalize confidence saturation directly.
- **Budget-aware evaluation:** report reliability as a function of query budget M and decoding constraints, enabling cost-sensitive comparison.
- **Domain-stratified analysis:** explicitly separate knowledge-intensive from reasoning-intensive subsets and quantify failure prediction gaps per category.

5.2 Elicitation formats that reduce discretization and stylistic bias

- **Structured confidence outputs:** replace single scalar percentages with calibrated bins, interval estimates, or calibrated rankings that are less susceptible to conversational conventions.
- **Consistency-anchored elicitation:** elicit confidence conditional on alternative answers or counterfactuals, forcing score variation that can support failure prediction rather than constant high confidence.

5.3 Hybrid black-box and limited white-box integration

The paper notes that some APIs provide partial internal signals such as logits. A concrete direction is to combine black-box elicitation (verbalized confidence, self-probing) with limited white-box features (token-level probabilities for answer spans) in a constrained setting, evaluating whether small amounts of internal access disproportionately improve failure prediction for knowledge-intensive tasks.

5.4 Semantics-aware uncertainty estimation under resource constraints

The white-box token-probability baselines remain weak at capturing semantic uncertainty. A promising direction is to develop lightweight semantic-consistency signals that approximate meaning-level uncertainty without full semantic uncertainty pipelines that are computationally expensive. Candidate approaches include paraphrase-invariant agreement tests, entailment-style checks using compact verifier models, or retrieval-grounded consistency verification, evaluated under strict cost budgets.

5.5 Adaptive sampling and stopping rules

Since performance improves and then saturates with larger M , research can target adaptive policies that decide when to stop sampling:

- **Early stopping on stability:** stop when answer distribution concentrates beyond a threshold.
- **Uncertainty-triggered escalation:** increase M only when initial samples disagree or when confidence signals indicate risk.

Such policies can preserve the failure prediction benefits of variance while controlling query costs.

5.6 Understanding and correcting overconfidence mechanisms

The observed preference for multiples of 5 and high-confidence saturation suggests systematic generation biases. Research can focus on:

- **Prompt-level regularization:** prompts that enforce score dispersion when evidence is weak.
- **Post-hoc calibration layers:** external calibrators trained on held-out correctness labels, operating on elicited features such as confidence text, rationales, and agreement statistics, while preserving black-box constraints.

6 Conclusion

The study provides a well-structured empirical framework for black-box confidence elicitation in LLMs, with methodological strengths including clear decomposition into prompting, sampling, and aggregation components, evaluation on orthogonal objectives (calibration and failure prediction), broad benchmark and model coverage, and controlled component-wise analyses. Empirical results establish that vanilla verbalized confidence is substantially overconfident (high confidence mass in 80–100 and discretized values) and often weak at distinguishing correct from incorrect responses. Scaling model capability improves both ECE and AUROC, but performance remains far from deployment-ready on challenging tasks, especially those requiring specialized knowledge. Sampling-driven variance and carefully chosen aggregation can substantially improve failure prediction in several regimes, but these gains incur linear query costs and do not universally resolve error detection.

The most critical limitations arise from restricted task scope (single-ground-truth QA), incomplete cost and hyperparameter reporting for full operational reproducibility, and structural misalignment between calibration gains and per-instance failure prediction. The main technical bottlenecks include the difficulty of capturing semantic uncertainty through black-box textual channels, the dependence of variance signals on expensive sampling, and instability or assumption sensitivity in aggregation mechanisms. The most actionable research directions involve objective-aligned, budget-aware evaluation protocols; elicitation formats that reduce discretization and induce informative score variation; hybrid strategies that exploit limited internal access when available; semantics-aware verification signals that remain cost-effective; and adaptive sampling policies that trade cost for reliability in a principled manner.